

An oscillator does not create energy. It merely acts as an energy converter, drawing power from the DC power supply and converting it into oscillations.

The basic circuit to generate electrical oscillations is the combination of an inductor and capacitor, also known as tank circuit. The capacitor stores energy in the electrostatic field, whereas the inductor stores energy in the electromagnetic field. Energy transformation between the electrostatic and electromagnetic fields results in oscillations. The L-C combination, shown in Fig. 1, controls the oscillation. The initial starting signal is due to stray pick-up by the capacitor.

Let capacitor C be charged by some stray DC source. On closing the switch, the capacitor discharges through inductor L. Electrons from the negative plate flow in the direction shown in Fig. 1(b). The flow of electrons is actually the current through the inductor, which results in building up of a magnetic field around the inductor. Its magnitude will be maximum when the capacitor is fully discharged. In this manner, the electrostatic energy of the capacitor is converted into electromagnetic energy.

Now it is the turn of the magnetic field to act. Since there is no current as the capacitor has fully discharged, the magnetic field begins to decay, producing an induced e.m.f. Thereby the capacitor begins to charge but with opposite polarity (see Fig. 1(c)). It attains maximum charge when the magnetic field reduces to zero. The energy stored in the magnetic field is thus converted into electrostatic energy.

The capacitor is again set to discharge through the coil, resulting in the build-up of a magnetic field (see Fig. 1(d)). But the direction of flow of electrons is opposite and so is the direction of the magnetic field.

The cycle repeats, with the sequence of charge and discharge of the capacitor resulting in alternate

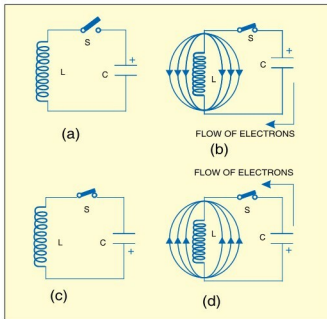


Fig. 1: Charging and discharging of a capacitor through an inductor

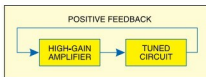


Fig. 2: Block diagram of an oscillator

motion of electrons or oscillating current. The energy is alternately stored in the form of electrostatic (capacitor) and magnetic fields (coil). This interchange of energy between L and C is repeated over and over again, resulting in the generation of electrical oscillations.

The amplitude of oscillations, however, gradually decreases. This is because during each cycle, a part of the energy is lost in the form of heat due to resistance of the coil. For practical applications, sustained oscillations can be achieved by supplying a proper amount of energy at proper phase.

The oscillator circuit is realised with a high-gain amplifier, which generates an input signal for itself through feedback. It is a case of positive feedback where a part of the output is fed back in phase for building of oscillations and then sustaining them. The arrangement is shown in

Fig. 2.

The frequency of an oscillator is controlled by means of tuned, or resonant, circuits. These are usually inductor-capacitor (L-C), crystal or resistor-capacitor (R-C) combinations. The L-C scheme is common at RF, while the R-C method is used more for audio oscillators.

The tuned circuit makes the feedback path favourable for the

resonant frequency but faces a high-impedance path at frequencies other than the resonant frequency. As a result, oscillations take place at a predictable and stable frequency, determined by the inductance and capacitance or by the resistance and capacitance.

There are many types of oscillators, common ones being Hartley, Colpitts, Clapp, Armstrong, Wein Bridge and R-C phase-shift. For circuit diagram, operation and analysis of these oscillators, you may refer to any book on electronic circuits.

Coming to applications, oscillators are used in radio communications. These generate a carrier frequency, which is modulated by the modulating signal. Oscillators are also used in radio and TV receivers for frequency control and for mixing and detection. Audio-frequency oscillators find use in music synthesisers, fax modems, doorbells, beepers, sirens and alarms, and electronic toys.

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